From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

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Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer using SLURM for job scheduling (Yoo et al., 2003), GNU Parallel to run the simulations in parallel (Tange, 2021), and Apptainer to ensure a reproducible software stack (Kurtzer et al., 2017, 2021).

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Links

- **Research compendium:** OSF, GitHub, and the compendium website.
- **R packages:** `fitVARMxID` for person-specific estimation and `metaDyn` for meta-analytic synthesis; source code is available at `fitVARMxID` on GitHub and `metaDyn` on GitHub.
- **Worked examples:** one simulation replication, including Mplus inputs, priors, and diagnostics; the empirical analysis; and the computational benchmark.
- **Reproduction:** container instructions.

Additional Information

Prior Sensitivity

The simulation compares dynamic structural equation modeling (DSEM) under default and alternative priors. Defaults use diffuse normal location priors and IW($\mathbf{0}, -d-1$) for covariance blocks of dimension d (Asparouhov & Muthén, 2023). Alternatives use $N(0, 25)$ for set-point means, $N(0, 0.25)$ for transition means, IW($\mathbf{I}_2, 3$) for set-point and innovation covariances, and IW($0.004\mathbf{I}_4, 5$) for transition covariance. Normal-prior second arguments are variances. The alternative transition-variance marginals have mode .001 and no finite mean; these proper, heavy-tailed priors assess sensitivity near small variances rather than eliminate prior influence. The alternative priors are not centered on the generating values. Priors centered on the true values would give DSEM population information that is unavailable in practice and that MetaVAR does not use, so the comparison would partly measure prior information rather than estimator performance. Data-informed priors are examined in the empirical comparison (Section). The simulation vignette provides example inputs and prior printouts.

Usable Replications and Boundary Estimates

Table 1 reports usable replications and boundary frequencies for every condition. MetaVAR estimates $\boldsymbol{\tau}^2 = \mathbf{LDL}'$ using unit lower-triangular $\mathbf{L}$ and softplus-transformed diagonal entries plus 10^{-8} in $\mathbf{D}$. Otherwise usable near-boundary fits are retained. Near-zero variances and near-singular covariance matrices are distinct from inadmissible fits; neither positive parameterization nor robust Wald intervals guarantees valid inference at zero. Figure 6 reports zero-target exclusion separately from nonzero detection. Bayesian variance intervals exclude zero by construction.

Table 2 compares DSEM random-effect failure rates under the strict criterion of Table 1 and the moderate criterion ($\widehat{R} \leq 1.05$, bulk and tail effective sample sizes ≥ 100, relative Monte Carlo error ≤ 0.10). Table 3 reports practical-homogeneity results for the transition variances in the zero-heterogeneity cells at three tolerances.

Empirical Comparison

MetaVAR and the uncertainty-uncorrected benchmark (Uncorr) use identical nine-feature Stage 1 estimates for the 212 retained participants. MetaVAR also uses the full joint sampling covariance; a positive-definite approximation was applied when its smallest eigenvalue was at or below 10^{-6} (43 participants; maximum entry change 2.40×10^{-5}). We fitted DSEM to the same 212 participants under four prior specifications in `Mplus` 9.1, each with four chains of 40,000 iterations; the first half of each chain was discarded as burn-in. The default fit left priors to `Mplus`. The alternative fit used $N(3, 4)$ for set-point means, $N(0, 0.25)$ for transition means, $N(-1, 1)$ for the means of the log innovation variances, and $N(0, 0.5)$ for the mean Fisher-z innovation correlation, with IW($\mathbf{I}_2, 3$), IW($0.25\mathbf{I}_4, 5$), and IW($\mathbf{I}_3, 4$) for the set-point, transition, and innovation covariance blocks. The two data-informed fits, which follow the recommendation to include a data-based prior in sensitivity analyses of inverse-Wishart specifications (Schuurman et al., 2016), centered every prior on the MetaVAR estimates. Each mean had the prior $N(\hat{\alpha}_j, \widehat{\mathrm{Var}}(\hat{\alpha}_j))$, with the MetaVAR estimate and its sampling variance. Each covariance block b of dimension d_b had the prior IW($n\hat{\boldsymbol{\tau}}_b^2$, $n + d_b + 1$) with $n = 212$, whose mean is the MetaVAR estimate $\hat{\boldsymbol{\tau}}_b^2$; the prior therefore carries the weight of 212 persons. The full version used this scale matrix; the diagonal version used $n\,\mathrm{diag}(\hat{\boldsymbol{\tau}}_b^2)$, so its prior mean kept the MetaVAR variances and set the covariances to zero. Because DSEM parameterizes the innovation covariance by log variances and a Fisher-z correlation, the MetaVAR innovation estimates and their covariance were mapped to that scale by the delta method.~~Both empirical DSEM fits terminated normally but had inadequate posterior convergence: rank-normalized $\widehat{R}$ for the positive-affect inertia variance was 1.442 under default priors and 1.462 under alternatives, with bulk effective sample sizes below eight. Inertia means also failed the posterior-quality screen. These fits therefore cannot support a reliable numerical comparison; their saved output, full diagnostics, and trace plots remain in the compendium. The shared-input MetaVAR–Uncorr comparison in the manuscript illustrates the effect of uncertainty propagation, without establishing accuracy against unknown population values.~~

The default and alternative fits terminated normally but did not converge. Their final potential scale reduction values were 1.451 and 1.366, and rank-normalized $\widehat{R}$ exceeded 1.05 for 10 of the 28 parameters in each fit, all among the transition means, variances, and covariances; $\widehat{R}$ for the PA-inertia variance was 1.442 and 1.462, with bulk effective sample sizes below eight. The full data-informed fit converged (final potential scale reduction 1.023; $\widehat{R} \leq 1.03$ for every parameter). The diagonal fit nearly did (1.050); only the NA-inertia mean and variance had $\widehat{R}$ above 1.05 (both 1.06). The four fits took 36 hours, 3 minutes; 35 hours, 36 minutes; 35 hours, 19 minutes; and 35 hours, 28 minutes. In every fit, `Mplus` removed posterior draws that implied non-stationary person-specific dynamics for 6 to 10 participants. It also reported a correlation of 1.00 between the mean Fisher-z innovation correlation and the median

innovation correlation; the second is the hyperbolic tangent of the first, so this correlation is expected and does not indicate poor identification.

Table 4 compares the estimates. Set-point means and variances and the mean innovation correlation agreed closely across the four fits. The variances of the log innovation variances were larger in every DSEM fit than in MetaVAR's delta-method values, with the data-informed fits in between; this comparison is approximate because the two methods parameterize the innovation covariance differently. The weak-prior fits gave higher inertia means and much smaller transition variances than MetaVAR. The data-informed priors moved the transition means close to the MetaVAR values and the transition variances most of the way. The transition covariances followed the prior's scale matrix: they approached the MetaVAR values under the full version, but stayed near zero under the diagonal version, as in the weak-prior fits. These patterns suggest that the data weakly identify between-person heterogeneity in the dynamics, especially its covariances. The data-informed fits reuse the data at a weight equal to the sample, so their agreement with MetaVAR and their smaller posterior standard deviations (by a factor close to $1/\sqrt{2}$ for the set-point means) arise partly by construction; they are a sensitivity analysis, not independent evidence about which estimator is more accurate.

Simulation Performance Figures

Indices 1–6 denote $(\mu_1, \mu_2, a_{11}, a_{21}, a_{12}, a_{22})$; α denotes population means and τ^2 between-person variances/covariances. Figures include all 36 conditions and both DSEM priors. Unbalanced denotes cyclic lengths of 50, 100, and 200. Recovery uses calibrated targets after stability rejection. MetaVAR uses normal-theory intervals. Relative bias and detection apply to nonzero targets; zero targets use absolute bias and separately displayed interval exclusion. Grey tiles denote undefined or unavailable metrics.

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Table 1

Usable Replications, MetaVAR Boundaries, and DSEM Posterior Quality

Cell	h	N	T	Meta usable	Near singular	Near zero	DSEM-D pass	DSEM-A pass
1	1	50	50	1000	0.635	0.000	477	0
2	1	100	50	1000	0.323	0.000	95	0
3	1	200	50	1000	0.092	0.000	0	0
4	1	50	100	1000	0.629	0.000	610	7
5	1	100	100	1000	0.373	0.000	242	2
6	1	200	100	999	0.183	0.000	58	0
7	1	50	200	1000	0.474	0.000	869	508
8	1	100	200	1000	0.262	0.000	681	476
9	1	200	200	1000	0.132	0.000	424	348
10	1	50	U	1000	0.618	0.000	448	7
11	1	100	U	1000	0.352	0.000	218	10
12	1	200	U	1000	0.154	0.000	71	6
13	2	50	50	999	0.483	0.000	562	0
14	2	100	50	1000	0.195	0.000	173	0
15	2	200	50	1000	0.047	0.000	22	0
16	2	50	100	1000	0.445	0.000	716	42
17	2	100	100	1000	0.196	0.001	450	27
18	2	200	100	1000	0.073	0.000	231	7
19	2	50	200	1000	0.259	0.000	910	673
20	2	100	200	1000	0.077	0.000	834	589
21	2	200	200	1000	0.014	0.000	750	429
22	2	50	U	1000	0.453	0.000	546	48
23	2	100	U	1000	0.202	0.000	379	59
24	2	200	U	1000	0.042	0.000	266	23
25	0	50	50	996	0.980	0.006	138	0
26	0	100	50	997	0.936	0.003	2	0
27	0	200	50	998	0.797	0.001	0	0
28	0	50	100	991	0.998	0.015	205	0
29	0	100	100	992	0.989	0.007	6	0
30	0	200	100	995	0.959	0.002	0	0
31	0	50	200	983	0.996	0.020	413	0
32	0	100	200	990	0.995	0.026	41	0
33	0	200	200	982	0.993	0.013	0	1
34	0	50	U	982	0.996	0.015	42	0
35	0	100	U	992	0.989	0.016	0	0
36	0	200	U	986	0.982	0.009	0	0

Note. Each cell requests 1,000 replications. All Stage 1, Uncorr, and both DSEM branches have 1,000 recorded usable outputs; MetaVAR retains 35,882 of 36,000 (118 inadmissible). U denotes cyclic unequal lengths; h is the transition-covariance multiplier. DSEM-D and DSEM-A denote default and alternative priors. Near singular means minimum covariance eigenvalue below 10^{-8}; near zero means a diagonal variance at most 10^{-6}. These rates condition on retained fits and are distinct from failures. DSEM passing counts require every diagnostic to be finite, $\widehat{R} \leq 1.01$, bulk and tail effective sample sizes ≥ 400, and Monte Carlo error relative to posterior SD ≤ 0.05. They do not replace primary denominators.

Table 2

DSEM Random-Effect Failure Rates Under Strict and Moderate Posterior-Quality Criteria

				DSEM-D		DSEM-A	
Cell	h	N	T	Strict	Moderate	Strict	Moderate
1	1	50	50	.521	.000	1.000	.472
2	1	100	50	.904	.012	1.000	.637
3	1	200	50	1.000	.149	1.000	.834
4	1	50	100	.389	.000	.993	.013
5	1	100	100	.755	.004	.998	.020
6	1	200	100	.942	.015	1.000	.040
7	1	50	200	.131	.000	.492	.000
8	1	100	200	.319	.001	.524	.000
9	1	200	200	.576	.000	.652	.000
10	1	50	U	.550	.000	.993	.009
11	1	100	U	.781	.001	.990	.004
12	1	200	U	.929	.006	.994	.007
13	2	50	50	.435	.001	1.000	.336
14	2	100	50	.824	.009	1.000	.502
15	2	200	50	.978	.100	1.000	.678
16	2	50	100	.283	.000	.958	.002
17	2	100	100	.547	.004	.973	.002
18	2	200	100	.769	.006	.993	.025
19	2	50	200	.090	.000	.327	.000
20	2	100	200	.165	.001	.411	.000
21	2	200	200	.250	.000	.571	.000
22	2	50	U	.450	.001	.952	.002
23	2	100	U	.620	.000	.941	.001
24	2	200	U	.734	.002	.977	.005
25	0	50	50	.862	.000	1.000	.965
26	0	100	50	.998	.003	1.000	.999
27	0	200	50	1.000	.135	1.000	1.000
28	0	50	100	.795	.000	1.000	.118
29	0	100	100	.994	.000	1.000	.340
30	0	200	100	1.000	.010	1.000	.754
31	0	50	200	.587	.000	1.000	.000
32	0	100	200	.959	.000	1.000	.000
33	0	200	200	1.000	.000	.999	.000
34	0	50	U	.958	.000	1.000	.090
35	0	100	U	1.000	.000	1.000	.189
36	0	200	U	1.000	.018	1.000	.410

Note. Entries are proportions of replications in which at least one random-effect (variance or covariance) parameter failed the criterion. Strict: $\widehat{R} \leq 1.01$, bulk and tail effective sample sizes ≥ 400, Monte Carlo error relative to posterior SD ≤ 0.05. Moderate: $\widehat{R} \leq 1.05$, effective sample sizes ≥ 100, relative Monte Carlo error ≤ 0.10. U denotes cyclic unequal lengths; h is the transition-covariance multiplier. DSEM-D and DSEM-A denote default and alternative priors.

Table 3

Practical Homogeneity of Transition Variances at Zero Heterogeneity

	$\delta = .0004$ (SD .02)		$\delta = .0009$ (SD .03)		$\delta = .0025$ (SD .05)	
Method	Upper $\leq \delta$	Lower $> \delta$	Upper $\leq \delta$	Lower $> \delta$	Upper $\leq \delta$	Lower $> \delta$
MetaVAR, normal	.029 [.000, .107]	.039 [.000, .313]	.093 [.001, .401]	.023 [.000, .237]	.364 [.016, .988]	.006 [.000, .074]
MetaVAR, robust	.042 [.000, .142]	.050 [.003, .314]	.118 [.003, .421]	.029 [.000, .232]	.396 [.024, .985]	.007 [.000, .090]
Uncorr	.000 [.000, .000]	1.000 [1.000, 1.000]	.000 [.000, .000]	1.000 [1.000, 1.000]	.000 [.000, .000]	.988 [.699, 1.000]
DSEM-D	.000 [.000, .000]	1.000 [1.000, 1.000]	.000 [.000, .000]	.761 [.040, 1.000]	.056 [.000, .906]	.062 [.000, .626]
DSEM-A	.000 [.000, .000]	1.000 [1.000, 1.000]	.000 [.000, .000]	.025 [.000, .195]	.262 [.000, .998]	.000 [.000, .000]

Note. Entries are mean proportions of replications over the 12 zero-heterogeneity cells and four transition variances, with the minimum and maximum over these 48 cell-by-variance combinations in brackets. Upper $\leq \delta$: the upper 95% limit is at or below the tolerance (practical homogeneity supported). Lower $> \delta$: the lower 95% limit exceeds the tolerance (heterogeneity beyond the tolerance concluded). MetaVAR and Uncorr use confidence intervals; DSEM uses posterior credible intervals. $\delta = .0009$ corresponds to a between-person SD of .03, the benchmark for a small cross-lagged effect (Orth et al., 2024).

Table 4

Empirical DSEM Estimates Under Four Prior Specifications, With MetaVAR

		Default		Alternative		Data-informed, full		Data-informed, diagonal	
Parameter	MetaVAR	Median (SD)	$\widehat{R}$	Median (SD)	$\widehat{R}$	Median (SD)	$\widehat{R}$	Median (SD)	$\widehat{R}$
Means									
μ_{NA}	1.442	1.440 (.026)	1.00	1.440 (.026)	1.00	1.441 (.018)	1.00	1.441 (.018)	1.00
μ_{PA}	2.573	2.574 (.033)	1.00	2.574 (.033)	1.00	2.573 (.023)	1.00	2.573 (.023)	1.00
$\mu_{a_{11}}$ (NA inertia)	.383	.467 (.029)	1.18	.467 (.025)	1.05	.401 (.022)	1.02	.398 (.023)	1.06
$\mu_{a_{21}}$ (NA→PA)	−.095	−.050 (.016)	1.01	−.058 (.020)	1.18	−.088 (.022)	1.01	−.092 (.023)	1.01
$\mu_{a_{12}}$ (PA→NA)	−.039	−.021 (.008)	1.18	−.029 (.010)	1.09	−.041 (.012)	1.00	−.044 (.013)	1.00
$\mu_{a_{22}}$ (PA inertia)	.329	.452 (.027)	1.23	.452 (.026)	1.19	.348 (.022)	1.01	.351 (.022)	1.01
$\mu \log \psi_{11}$	−3.448	−3.591 (.074)	1.00	−3.580 (.072)	1.00	−3.515 (.042)	1.00	−3.515 (.042)	1.00
$\mu \log \psi_{22}$	−2.776	−2.894 (.062)	1.00	−2.883 (.062)	1.00	−2.819 (.034)	1.00	−2.822 (.035)	1.00
$\mu\ z_{\psi_{21}}$	−.179	−.194 (.020)	1.01	−.191 (.021)	1.00	−.189 (.012)	1.00	−.189 (.013)	1.00
Set-point covariance									
τ^2_{NA}	.134	.138 (.014)	1.00	.139 (.014)	1.00	.134 (.009)	1.00	.134 (.009)	1.00
$\tau_{\mathrm{PA,NA}}$	.009	.009 (.013)	1.00	.009 (.012)	1.00	.009 (.008)	1.00	.004 (.008)	1.00
τ^2_{PA}	.219	.225 (.023)	1.00	.223 (.022)	1.00	.218 (.015)	1.00	.218 (.015)	1.00
Transition covariance									
$\tau^2_{a_{11}}$	.190	.095 (.018)	1.36	.087 (.014)	1.12	.153 (.012)	1.03	.153 (.012)	1.06
$\tau_{a_{21},a_{11}}$	−.030	−.008 (.006)	1.08	−.007 (.007)	1.26	−.022 (.008)	1.02	−.004 (.008)	1.02
$\tau^2_{a_{21}}$	.175	.015 (.004)	1.07	.020 (.004)	1.03	.127 (.010)	1.00	.127 (.010)	1.01
$\tau_{a_{12},a_{11}}$	.020	.002 (.003)	1.06	.001 (.003)	1.13	.013 (.004)	1.01	.001 (.004)	1.00
$\tau_{a_{12},a_{21}}$	.008	.004 (.001)	1.03	.003 (.002)	1.03	.008 (.004)	1.01	.004 (.004)	1.01
$\tau^2_{a_{12}}$	.058	.003 (.001)	1.00	.008 (.001)	1.01	.041 (.003)	1.00	.041 (.003)	1.00
$\tau_{a_{22},a_{11}}$	.002	.049 (.014)	1.45	.037 (.010)	1.10	.009 (.008)	1.00	.007 (.008)	1.00
$\tau_{a_{22},a_{21}}$	.041	.005 (.005)	1.02	.005 (.006)	1.22	.029 (.008)	1.00	.005 (.007)	1.01
$\tau_{a_{22},a_{12}}$	−.029	−.001 (.002)	1.11	−.002 (.003)	1.09	−.019 (.004)	1.00	−.002 (.004)	1.00
$\tau^2_{a_{22}}$	.189	.076 (.015)	1.44	.070 (.013)	1.46	.150 (.011)	1.01	.149 (.011)	1.02
Innovation covariance									
$\tau^2 \log \psi_{11}$	.599	1.093 (.114)	1.00	1.059 (.108)	1.00	.816 (.058)	1.00	.817 (.058)	1.00
$\tau \log \psi_{22}$, $\log \psi_{11}$	.172	.434 (.075)	1.00	.415 (.071)	1.00	.289 (.037)	1.00	.198 (.036)	1.00
$\tau^2 \log \psi_{22}$	.394	.786 (.082)	1.00	.760 (.077)	1.00	.570 (.041)	1.00	.571 (.041)	1.00
$\tau\ z_{\psi_{21}}$, $\log \psi_{11}$	−.056	−.068 (.023)	1.00	−.066 (.022)	1.00	−.065 (.011)	1.00	−.030 (.011)	1.00
$\tau\ z_{\psi_{21}}$, $\log \psi_{22}$	−.030	−.077 (.020)	1.00	−.074 (.019)	1.00	−.052 (.009)	1.00	−.033 (.009)	1.00
$\tau^2\ z_{\psi_{21}}$	.035	.069 (.008)	1.00	.073 (.009)	1.00	.046 (.004)	1.00	.046 (.004)	1.00

Note. DSEM entries are posterior medians and standard deviations (SD) from four chains of 40,000 iterations, after discarding the first half of each chain, with rank-normalized $\widehat{R}$. MetaVAR entries are the estimates that center the data-informed priors; the innovation block (log innovation variances $\log \psi_{11}$ and $\log \psi_{22}$ and the Fisher-z innovation correlation $z_{\psi_{21}}$) is mapped to the DSEM scale by the delta method. τ^2 denotes a between-person variance and τ a covariance.

Figure 1

Overall Performance

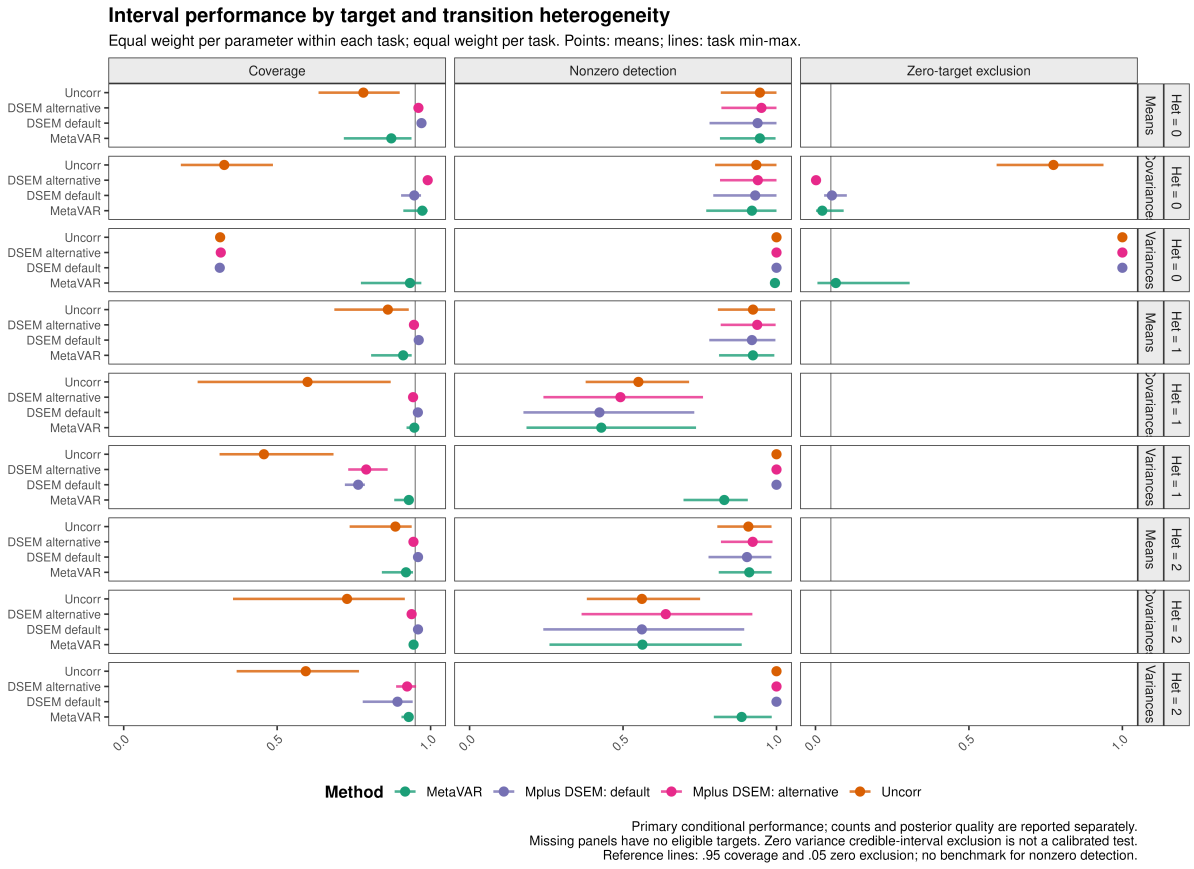


Figure 2

Coverage Across Transition-Heterogeneity Conditions

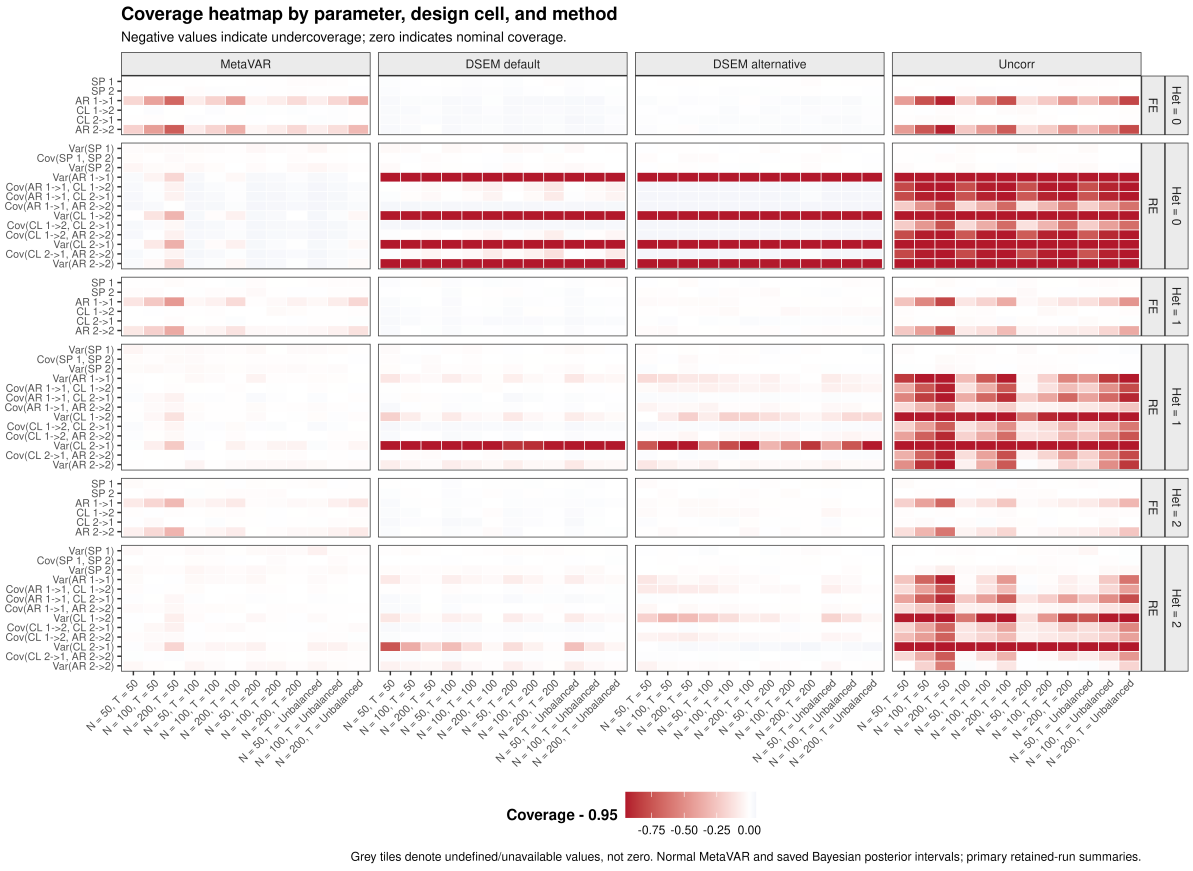
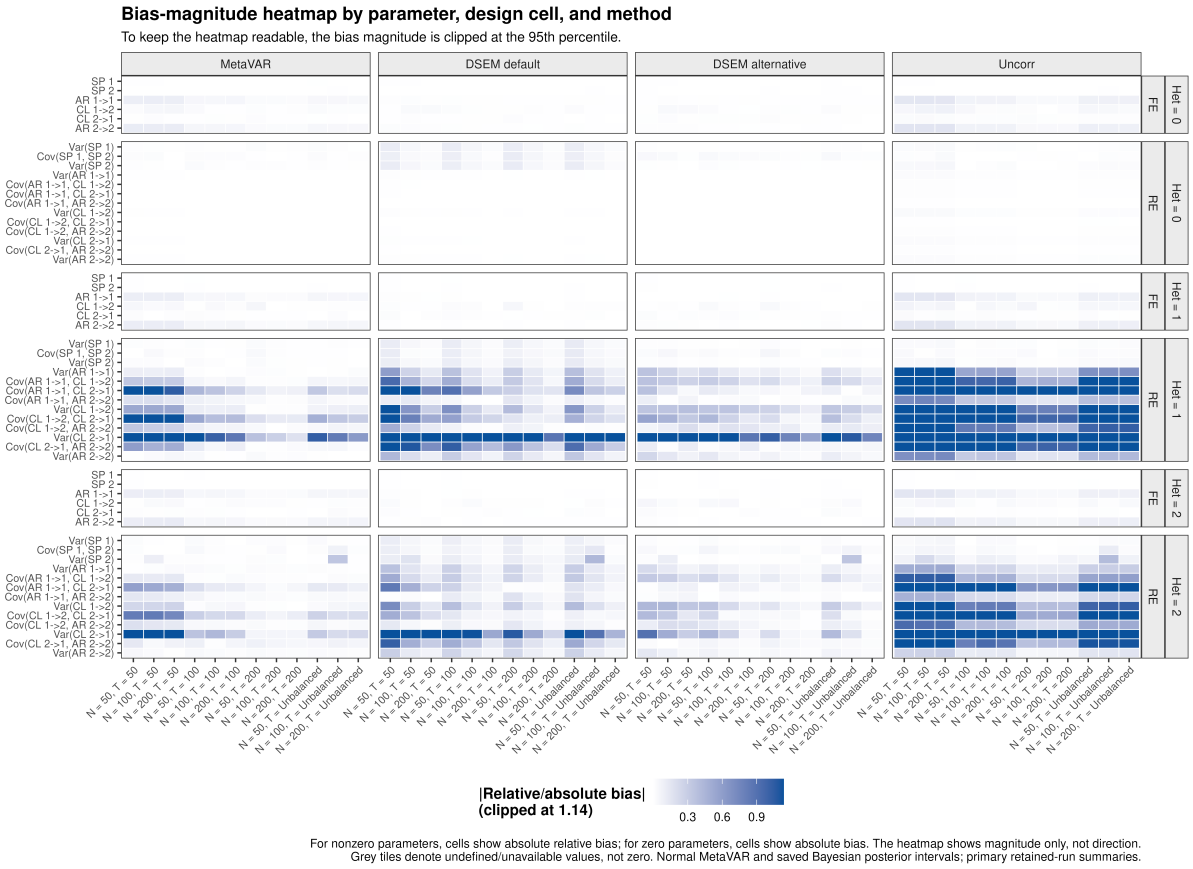


Figure 3

Absolute Relative or Absolute Bias Across Transition-Heterogeneity Conditions



For nonzero parameters, cells show absolute relative bias; for zero parameters, cells show absolute bias. The heatmap shows magnitude only, not direction. Grey tiles denote undefined/unavailable values, not zero. Normal MetaVAR and saved Bayesian posterior intervals; primary retained-run summaries.

Figure 4

Root Mean Square Error Across Transition-Heterogeneity Conditions

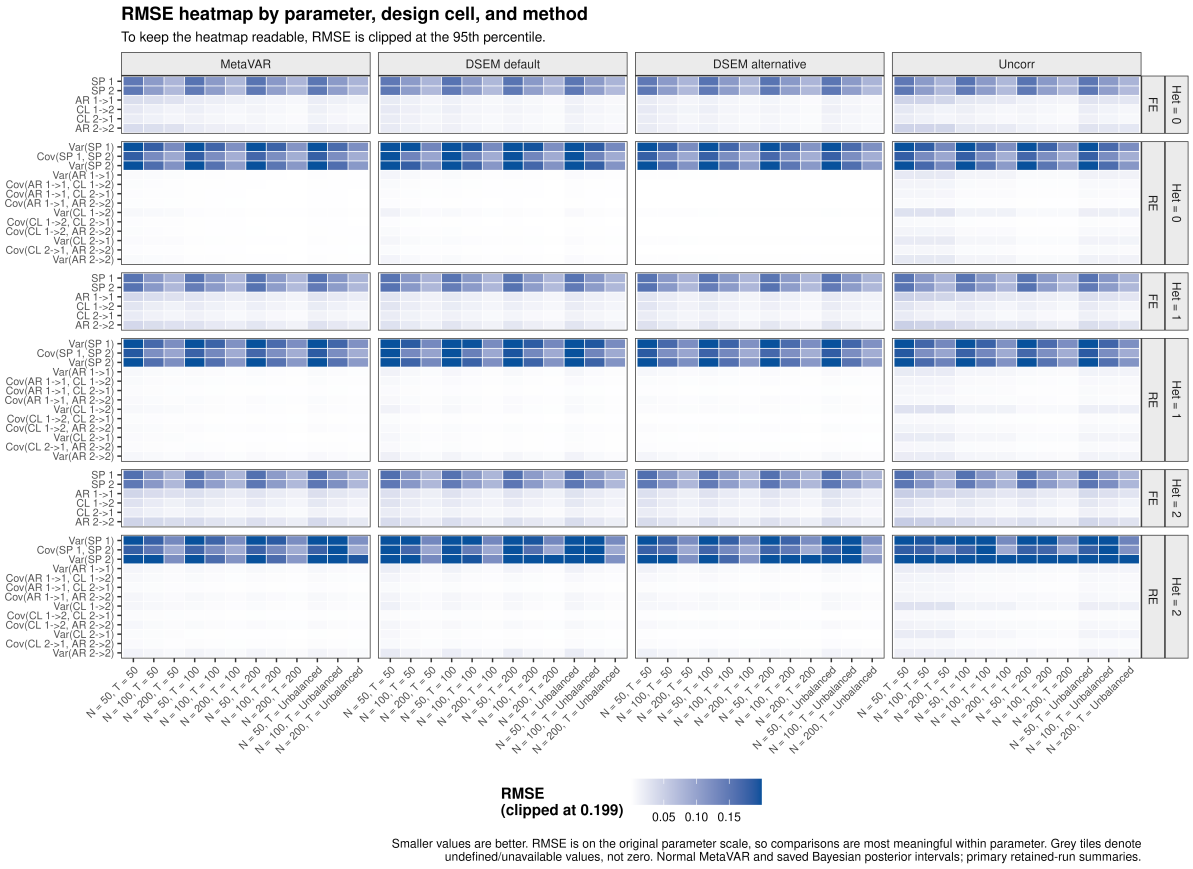


Figure 5

Detection of Nonzero Targets Across Transition-Heterogeneity Conditions

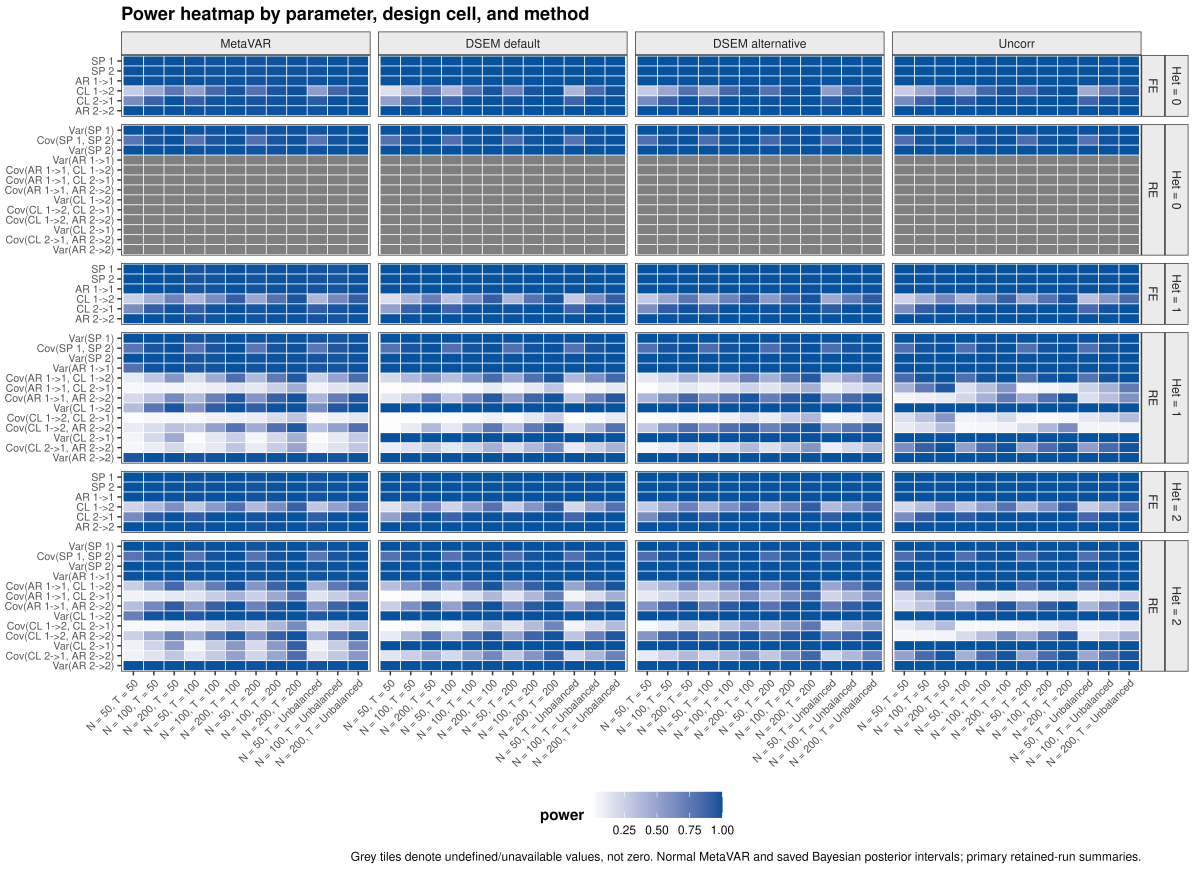


Figure 6
Interval Exclusion for Exact-Zero Targets (Bayesian Variance Intervals Exclude Zero by Construction)

